1)

Let $y = \operatorname{arsinh} x$

Then $\sinh y = x$

$\frac{\mathrm{d}}{\mathrm{d}x}(\sinh y) = \frac{\mathrm{d}}{\mathrm{d}x}(x)$

$\Rightarrow \cosh y \frac{\mathrm{d}y}{\mathrm{d}x} = 1$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\cosh y} = \frac{1}{\sqrt{\sinh^2 y + 1}} = \frac{1}{\sqrt{x^2+1}}$ as required

2)

Let $y = \operatorname{arcosh} x$

Then $\cosh y = x$

$\frac{\mathrm{d}}{\mathrm{d}x}(\cosh y) = \frac{\mathrm{d}}{\mathrm{d}x}(x)$

$\Rightarrow \sinh y \frac{\mathrm{d}y}{\mathrm{d}x} = 1$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\sinh y} = \frac{1}{\sqrt{\cosh^2 y - 1}} = \frac{1}{\sqrt{x^2-1}}$ as required

3)

Let $y = \operatorname{artanh} x$

Then $\tanh y = x$

$\frac{\mathrm{d}}{\mathrm{d}x}(\tanh y) = \frac{\mathrm{d}}{\mathrm{d}x}(x)$

$\Rightarrow \operatorname{sech}^2 y \frac{\mathrm{d}y}{\mathrm{d}x} = 1$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\operatorname{sech}^2 y} = \frac{1}{1 - \tanh^2 y} = \frac{1}{1-x^2}$ as required

4) a)

$y = \operatorname{arsinh} 3x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 3 \times \frac{1}{\sqrt{(3x)^2+1}} = \frac{3}{\sqrt{9x^2+1}}$

4) b)

$y = \operatorname{arcosh} 2x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 2 \times \frac{1}{\sqrt{(2x)^2-1}} = \frac{2}{\sqrt{4x^2-1}}$

4) c)

$y = \operatorname{artanh} 5x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 5 \times \frac{1}{1-(5x)^2} = \frac{5}{1-25x^2}$

4) d)

$y = \operatorname{arsinh} \frac{1}{2}x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{2} \times \frac{1}{\sqrt{\left(\frac{1}{2}x\right)^2+1}} = \frac{1}{2\sqrt{\frac{1}{4}x^2+1}} = \frac{1}{2\sqrt{\frac{1}{4}(x^2+4)}} = \frac{1}{\sqrt{x^2+4}}$

4) e)

$y = \operatorname{arcosh} \frac{x}{4}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{4} \times \frac{1}{\sqrt{\left(\frac{1}{4}x\right)^2-1}} = \frac{1}{4\sqrt{\frac{1}{16}x^2-1}} = \frac{1}{4\sqrt{\frac{1}{16}(x^2-16)}} = \frac{1}{\sqrt{x^2-16}}$

4) f)

$y = \operatorname{artanh}\left(-\frac{x}{6}\right)$

$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{6} \times \frac{1}{1-\left(-\frac{1}{6}x\right)^2} = -\frac{1}{6\left(1-\frac{1}{36}x^2\right)} = -\frac{1}{6-\frac{1}{6}x^2} = -\frac{6}{36-x^2}$

5) a)

$y = 2\operatorname{arsinh} 2x - 2\operatorname{arcosh} 4x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 2 \times 2 \times \frac{1}{\sqrt{(2x)^2+1}} - 2 \times 4 \times \frac{1}{\sqrt{(4x)^2-1}}$

$= \frac{4}{\sqrt{4x^2+1}} - \frac{8}{\sqrt{16x^2-1}}$

5) b)

$y = \frac{1}{3}\operatorname{artanh} 3x - \frac{2}{3}\operatorname{artanh} 6x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{3} \times 3 \times \frac{1}{1-(3x)^2} - \frac{2}{3} \times 6 \times \frac{1}{1-(6x)^2}$

$= \frac{1}{1-9x^2} - \frac{4}{1-36x^2} = \frac{(1-36x^2)-4(1-9x^2)}{(1-9x^2)(1-36x^2)} = -\frac{3}{(1-9x^2)(1-36x^2)}$

6)

$y = 4\,\text{arsinh}\,x$

$y|_{x=2} = 4\ln\left(2+\sqrt{2^2+1}\right) = 4\ln\left(2+\sqrt{5}\right)$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 4\times\frac{1}{\sqrt{x^2+1}} = \frac{4}{\sqrt{x^2+1}}$

$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=2} = \frac{4}{\sqrt{2^2+1}} = \frac{4}{\sqrt{5}} = \frac{4\sqrt{5}}{5}$

So the tangent will have equation $y - 4\ln\left(2+\sqrt{5}\right) = \frac{4\sqrt{5}}{5}(x-2)$

$\Rightarrow y = \frac{4\sqrt{5}}{5}x + 4\ln\left(2+\sqrt{5}\right) - \frac{8\sqrt{5}}{5}$

7)

$y = -6\,\text{arcosh}\,\frac{x}{2}$

$y|_{x=4} = -6\ln\left(\frac{4}{2}+\sqrt{\left(\frac{4}{2}\right)^2-1}\right) = -6\ln\left(2+\sqrt{3}\right)$

$\frac{\mathrm{d}y}{\mathrm{d}x} = -6\times\frac{1}{2}\times\frac{1}{\sqrt{\left(\frac{x}{2}\right)^2-1}} = -\frac{3}{\sqrt{\frac{1}{4}x^2-1}} = -\frac{3}{\sqrt{\frac{1}{4}(x^2-4)}} = -\frac{6}{\sqrt{x^2-4}}$

$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=4} = -\frac{6}{\sqrt{4^2-4}} = -\frac{6}{\sqrt{12}} = -\frac{6}{2\sqrt{3}} = -\sqrt{3}$

So the tangent will have equation $y + 6\ln\left(2+\sqrt{3}\right) = -\sqrt{3}(x-4)$

$\Rightarrow y = -\sqrt{3}x + 4\sqrt{3} - 6\ln\left(2+\sqrt{3}\right)$

8)

$y = 10\,\text{artanh}\,\frac{x}{5}$

$y|_{x=3} = 10\times\frac{1}{2}\ln\left(\frac{1+\frac{3}{5}}{1-\frac{3}{5}}\right) = 5\ln 4$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 10\times\frac{1}{5}\times\frac{1}{1-\left(\frac{x}{5}\right)^2} = \frac{2}{1-\frac{x^2}{25}} = \frac{50}{25-x^2}$

$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=3} = \frac{50}{25-3^2} = \frac{50}{16} = \frac{25}{8}$

So the tangent will have equation $y - 5\ln 4 = \frac{25}{8}(x-3)$

$\Rightarrow y = \frac{25}{8}x + 5\ln 4 - \frac{75}{8}$

9)

$y = \operatorname{arsinh} 4x - \operatorname{arcosh} 2x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 4 \times \frac{1}{\sqrt{(4x)^2+1}} - 2 \times \frac{1}{\sqrt{(2x)^2-1}}$

$= \frac{4}{\sqrt{16x^2+1}} - \frac{2}{\sqrt{4x^2-1}}$

$= 4(16x^2+1)^{-\frac{1}{2}} - 2(4x^2-1)^{-\frac{1}{2}}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 4 \times 32x \times -\frac{1}{2}(16x^2+1)^{-\frac{3}{2}} - 2 \times 8x \times -\frac{1}{2}(4x^2-1)^{-\frac{3}{2}}$

$= -64x(16x^2+1)^{-\frac{3}{2}} + 8x(4x^2-1)^{-\frac{3}{2}}$

$\left.\frac{\mathrm{d}^2y}{\mathrm{d}x^2}\right|_{x=2} = -64 \times 2 \times (16(2)^2+1)^{-\frac{3}{2}} + 8 \times 2 \times (4(2)^2-1)^{-\frac{3}{2}}$

$= -128 \times 65^{-\frac{3}{2}} + 16 \times 15^{-\frac{3}{2}}$

$= -\frac{128\sqrt{65}}{4225} + \frac{16\sqrt{15}}{225}$

$= 0.0311 \ldots > 0$

As $\left.\frac{\mathrm{d}^2y}{\mathrm{d}x^2}\right|_{x=2} > 0$ the curve is convex (concave upwards) at $x = 2$

10)

$y = \sqrt{x^2+1} + \operatorname{arsinh} x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 2x \times \frac{1}{2}(x^2+1)^{-\frac{1}{2}} + \frac{1}{\sqrt{x^2+1}}$

$= \frac{x}{\sqrt{x^2+1}} + \frac{1}{\sqrt{x^2+1}}$

$= \frac{x+1}{\sqrt{x^2+1}}$

Stationary points exist when $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$

$\therefore \frac{x+1}{\sqrt{x^2+1}} = 0$

$\Rightarrow x + 1 = 0$

$\Rightarrow x = -1$

$y|_{x=-1} = \sqrt{(-1)^2+1} + \operatorname{arsinh}(-1)$

$= \sqrt{2} + \ln\left(-1 + \sqrt{(-1)^2+1}\right)$

$= \sqrt{2} + \ln\left(-1+\sqrt{2}\right)$

So the stationary point is at $\left(-1\,, \sqrt{2} + \ln\left(-1+\sqrt{2}\right)\right)$

11)

$y = 12\operatorname{arcosh} x - 3x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 12 \times \frac{1}{\sqrt{x^2-1}} - 3 = \frac{12}{\sqrt{x^2-1}} - 3$

Stationary points exist when $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$

$\therefore \frac{12}{\sqrt{x^2-1}} - 3 = 0$

$\Rightarrow \frac{12}{\sqrt{x^2-1}} = 3$

$\Rightarrow 12 = 3\sqrt{x^2-1}$

$\Rightarrow 4 = \sqrt{x^2-1}$

$\Rightarrow x^2 - 1 = 16$

$\Rightarrow x^2 = 17$

$\Rightarrow x = \sqrt{17}$ only as $\operatorname{arcosh} x$ only exists for $x \geq 1$

$\therefore y|_{x=\sqrt{17}} = 12\operatorname{arcosh}\left(\sqrt{17}\right) - 3\sqrt{17}$

$= 12\ln\left(\sqrt{17} + \sqrt{\left(\sqrt{17}\right)^2 - 1}\right) - 3\sqrt{17}$

$= 12\ln\left(4 + \sqrt{17}\right) - 3\sqrt{17}$

So the range is: $\left\{\mathrm{f}(x): \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \leq 12\ln\left(4 + \sqrt{17}\right) - 3\sqrt{17}\right\}$

12) a)

$A(3, 0)$

12) b)

$y = 2\sqrt{x^2 - 9} - 12\,\text{arcosh}\,\frac{x}{3}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 2 \times 2x \times \frac{1}{2}(x^2 - 9)^{-\frac{1}{2}} - 12 \times \frac{1}{3} \times \frac{1}{\sqrt{\left(\frac{x}{3}\right)^2 - 1}}$

$= \frac{2x}{\sqrt{x^2-9}} - \frac{4}{\sqrt{\left(\frac{x}{3}\right)^2 - 1}}$

$= \frac{2x}{\sqrt{x^2-9}} - \frac{4}{\sqrt{\frac{1}{9}(x^2-9)}}$

$= \frac{2x}{\sqrt{x^2-9}} - \frac{4}{\frac{1}{3}\sqrt{x^2-9}}$

$= \frac{2x}{\sqrt{x^2-9}} - \frac{12}{\sqrt{x^2-9}}$

$= \frac{2x-12}{\sqrt{x^2-9}}$

Stationary points exist when $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$

$\therefore \frac{2x-12}{\sqrt{x^2-9}} = 0$

$\Rightarrow 2x - 12 = 0$

$\Rightarrow x = 6$

$y|_{x=6} = 2\sqrt{6^2 - 9} - 12\,\text{arcosh}\,\frac{6}{3}$

$= 6\sqrt{3} - 12\ln\left(2 + \sqrt{2^2 - 1}\right)$

$= 6\sqrt{3} - 12\ln\left(2 + \sqrt{3}\right)$

So the stationary point is at $\left(6\,, 6\sqrt{3} - 12\ln\left(2 + \sqrt{3}\right)\right)$

So the range of f is:

$\left\{\text{f}(x): \text{f}(x) \in \mathbb{R}, \text{f}(x) \geq 6\sqrt{3} - 12\ln\left(2 + \sqrt{3}\right)\right\}$

13) a)

$y = 4\,\text{artanh}\,x + \ln(x+1) - 6x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{4}{1-x^2} + \frac{1}{x+1} - 6$

$= \frac{4}{(1-x)(1+x)} + \frac{1-x}{(1-x)(1+x)} - \frac{6(1-x)(1+x)}{(1-x)(1+x)}$

$= \frac{4+1-x-6(1-x^2)}{1-x^2}$

$= \frac{6x^2-x-1}{1-x^2}$

Stationary points exist when $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$

$\therefore \frac{6x^2-x-1}{1-x^2} = 0$

$\Rightarrow 6x^2 - x - 1 = 0$

$\Rightarrow (3x+1)(2x-1) = 0$

$\Rightarrow x = -\frac{1}{3}$ or $x = \frac{1}{2}$

$y|_{x=-\frac{1}{3}} = 4\,\text{artanh}\left(-\frac{1}{3}\right) + \ln\left(-\frac{1}{3}+1\right) - 6\left(-\frac{1}{3}\right)$

$= 4 \times \frac{1}{2}\ln\left(\frac{1+\left(-\frac{1}{3}\right)}{1-\left(-\frac{1}{3}\right)}\right) + \ln\frac{2}{3} + 2$

$= 2\ln\frac{1}{2} + \ln\frac{2}{3} + 2$

$= \ln\frac{1}{4} + \ln\frac{2}{3} + 2$

$= 2 + \ln\frac{1}{6}$

$= 2 - \ln 6$

$y|_{x=\frac{1}{2}} = 4\,\text{artanh}\left(\frac{1}{2}\right) + \ln\left(\frac{1}{2}+1\right) - 6\left(\frac{1}{2}\right)$

$= 4 \times \frac{1}{2}\ln\left(\frac{1+\frac{1}{2}}{1-\frac{1}{2}}\right) + \ln\frac{3}{2} - 3$

$= 2\ln 3 + \ln\frac{3}{2} - 3$

$= \ln 9 + \ln\frac{3}{2} - 3$

$= \ln\frac{27}{2} - 3$

So the stationary points are at $\left(-\frac{1}{3}, 2 - \ln 6\right)$ and $\left(\frac{1}{2}, \ln\frac{27}{2} - 3\right)$

13) b)

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{6x^2-x-1}{1-x^2}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{(1-x^2)\times(12x-1)-(6x^2-x-1)\times-2x}{(1-x^2)^2}$

$= \frac{(-12x^3+x^2+12x-1)-(-12x^3+2x^2+2x)}{(1-x^2)^2}$

$= \frac{-x^2+10x-1}{(1-x^2)^2}$

Points of inflection exist when $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = 0$

$\therefore \frac{-x^2+10x-1}{(1-x^2)^2} = 0$

$\Rightarrow -x^2 + 10x - 1 = 0$

$\Rightarrow x^2 - 10x + 1 = 0$

$\Rightarrow x = 5 \pm 2\sqrt{6}$

$5 + 2\sqrt{6} = 9.89$ (3 SF) and $5 - 2\sqrt{6} = 0.101$ (3 SF)

The domain of artanh x is $\{x: x \in \mathbb{R}, -1 < x < 1\}$ so $x \neq 5 + 2\sqrt{6}$

So we only need to check whether the 2nd derivative changes sign either side of $x = 5 - 2\sqrt{6}$.

$\left.\frac{\mathrm{d}^2y}{\mathrm{d}x^2}\right|_{x=0.1} = \frac{-(0.1)^2+10(0.1)-1}{(1-(0.1)^2)^2} = -0.0102 \ldots < 0$ (concave (concave downwards))

$\left.\frac{\mathrm{d}^2y}{\mathrm{d}x^2}\right|_{x=0.2} = \frac{-(0.2)^2+10(0.2)-1}{(1-(0.2)^2)^2} = 1.04 \ldots > 0$ (convex (concave upwards))

$\therefore$ There is a point of inflection for the curve at $x = 5 - 2\sqrt{6}$

14) a)

$y = \operatorname{arsinh}\left(\sqrt{x}\right) = \operatorname{arsinh}\left(x^{\frac{1}{2}}\right)$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{2}x^{-\frac{1}{2}} \times \frac{1}{\sqrt{\left(x^{\frac{1}{2}}\right)^2+1}} = \frac{1}{2\sqrt{x}} \times \frac{1}{\sqrt{x+1}} = \frac{1}{2\sqrt{x(x+1)}}$

14) b)

$y = \operatorname{arcosh}\left(\frac{1}{\sqrt{x}}\right) = \operatorname{arcosh}\left(x^{-\frac{1}{2}}\right)$

$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{2}x^{-\frac{3}{2}} \times \frac{1}{\sqrt{\left(x^{-\frac{1}{2}}\right)^2-1}} = -\frac{1}{2x^{\frac{3}{2}}} \times \frac{1}{\sqrt{x^{-1}-1}} = -\frac{1}{2\sqrt{x^3}} \times \frac{1}{\sqrt{x^{-1}-1}} = -\frac{1}{2\sqrt{x^2(1-x)}}$

14) c)

$y = \operatorname{artanh}\left(\sqrt[3]{x}\right) = \operatorname{artanh}\left(x^{\frac{1}{3}}\right)$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{3}x^{-\frac{2}{3}} \times \frac{1}{1-\left(x^{\frac{1}{3}}\right)^2} = \frac{1}{3x^{\frac{2}{3}}} \times \frac{1}{1-x^{\frac{2}{3}}} = \frac{1}{3x^{\frac{2}{3}}\left(1-x^{\frac{2}{3}}\right)}$

15) a)

$\frac{\mathrm{d}}{\mathrm{d}x}\left(\operatorname{arsinh}\frac{1}{x}\right) = \frac{\mathrm{d}}{\mathrm{d}x}\left(\operatorname{arsinh}x^{-1}\right) = -x^{-2} \times \frac{1}{\sqrt{(x^{-1})^2+1}}$

$= -\frac{1}{x^2} \times \frac{1}{\sqrt{\frac{1}{x^2}+1}} = -\frac{1}{x^2\sqrt{\frac{1}{x^2}(1+x^2)}} = -\frac{1}{x^2 \times \frac{1}{x} \times \sqrt{x^2+1}} = -\frac{1}{x\sqrt{x^2+1}}$ as required

15) b)

$y = \sqrt{x^2+1} - \operatorname{arsinh}\frac{1}{x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 2x \times \frac{1}{2}(x^2+1)^{-\frac{1}{2}} + \frac{1}{x\sqrt{x^2+1}}$

$= \frac{x}{\sqrt{x^2+1}} + \frac{1}{x\sqrt{x^2+1}} = \frac{x^2}{x\sqrt{x^2+1}} + \frac{1}{x\sqrt{x^2+1}} = \frac{x^2+1}{x\sqrt{x^2+1}}$

Stationary points exist when $\frac{\mathrm{d}y}{\mathrm{d}x} = 0$

$x^2+1 > 1$ for all $x > 0$, so there are no real values of x for which $\frac{x^2+1}{x\sqrt{x^2+1}} = 0$

Hence the curve has no stationary points

16) a)

$\frac{\mathrm{d}}{\mathrm{d}x}\left(\operatorname{artanh}\frac{2x}{\sqrt{4x^2+1}}\right) = \frac{\mathrm{d}}{\mathrm{d}x}\left(\operatorname{artanh}\frac{2x}{(4x^2+1)^{\frac{1}{2}}}\right)$

$= \frac{(4x^2+1)^{\frac{1}{2}} \times 2 - 2x \times 8x \times \frac{1}{2}(4x^2+1)^{-\frac{1}{2}}}{4x^2+1} \times \frac{1}{1-\left(\frac{2x}{(4x^2+1)^{\frac{1}{2}}}\right)^2}$

$= \frac{2(4x^2+1)^{\frac{1}{2}} - 8x^2(4x^2+1)^{-\frac{1}{2}}}{4x^2+1} \times \frac{1}{1-\frac{4x^2}{4x^2+1}}$

$= \frac{2(4x^2+1)-8x^2}{(4x^2+1)^{\frac{3}{2}}} \times \frac{4x^2+1}{4x^2+1-4x^2}$

$= \frac{2}{(4x^2+1)^{\frac{3}{2}}} \times \frac{4x^2+1}{1} = \frac{2}{(4x^2+1)^{\frac{1}{2}}} = \frac{2}{\sqrt{4x^2+1}}$ as required

16) b)

$y = \sqrt{4x^2+1} - \operatorname{artanh}\frac{2x}{\sqrt{4x^2+1}}$

$\frac{dy}{dx} = 8x \times \frac{1}{2}(4x^2+1)^{-\frac{1}{2}} - \frac{2}{\sqrt{4x^2+1}}$

$= \frac{4x}{\sqrt{4x^2+1}} - \frac{2}{\sqrt{4x^2+1}}$

$= \frac{4x-2}{\sqrt{4x^2+1}}$

Stationary points exist when $\frac{dy}{dx} = 0$

$\frac{4x-2}{\sqrt{4x^2+1}} = 0 \Rightarrow 4x - 2 = 0 \Rightarrow x = \frac{1}{2}$

$y|_{x=\frac{1}{2}} = \sqrt{4\left(\frac{1}{2}\right)^2+1} - \operatorname{artanh}\frac{2\left(\frac{1}{2}\right)}{\sqrt{4\left(\frac{1}{2}\right)^2+1}}$

$= \sqrt{2} - \operatorname{artanh}\left(\frac{\sqrt{2}}{2}\right)$

$= \sqrt{2} - \frac{1}{2}\ln\left(\frac{1+\frac{\sqrt{2}}{2}}{1-\frac{\sqrt{2}}{2}}\right)$

$= \sqrt{2} - \frac{1}{2}\ln\left(3+2\sqrt{2}\right)$

So there is a single stationary point at $\left(\frac{1}{2}, \sqrt{2} - \frac{1}{2}\ln\left(3+2\sqrt{2}\right)\right)$

17)

$\mathrm{f}(x) = \operatorname{arcosh} 2x + \operatorname{arcosh}(2x^{-1})$

$\mathrm{f}'(x) = \frac{2}{\sqrt{(2x)^2-1}} - 2x^{-2} \times \frac{1}{\sqrt{(2x^{-1})^2-1}}$

$= \frac{2}{\sqrt{4x^2-1}} - \frac{2}{x^2\sqrt{\frac{4}{x^2}-1}}$

$= \frac{2}{\sqrt{4x^2-1}} - \frac{2}{x^2\sqrt{\frac{1}{x^2}(4-x^2)}}$

$= \frac{2}{\sqrt{4x^2-1}} - \frac{2}{x\sqrt{4-x^2}}$

Stationary points exist when $\mathrm{f}'(x) = 0$

$\frac{2}{\sqrt{4x^2-1}} - \frac{2}{x\sqrt{4-x^2}} = 0$

$\Rightarrow \frac{1}{\sqrt{4x^2-1}} = \frac{1}{x\sqrt{4-x^2}}$

$\Rightarrow x\sqrt{4-x^2} = \sqrt{4x^2-1}$

$\Rightarrow x^2(4-x^2) = 4x^2 - 1$

$\Rightarrow 4x^2 - x^4 = 4x^2 - 1$

$\Rightarrow x^4 = 1$

$\Rightarrow x = 1$ as $x > 0$

$\mathrm{f}(1) = \operatorname{arcosh} 2 + \operatorname{arcosh} 2 = 2\operatorname{arcosh} 2$

$= 2 \times \ln\left(2 + \sqrt{2^2-1}\right) = 2\ln\left(2+\sqrt{3}\right) = \ln\left(7+4\sqrt{3}\right)$

So there is a stationary point at $\left(1, \ln\left(7+4\sqrt{3}\right)\right)$

18) a)

$y = \operatorname{arsinh}(\cosh x)$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \sinh x \times \frac{1}{\sqrt{\cosh^2 x+1}} = \frac{\sinh x}{\sqrt{\cosh^2 x+1}}$

18) b)

$y = \operatorname{arcosh}(\tanh x)$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \operatorname{sech}^2 x \times \frac{1}{\sqrt{\tanh^2 x-1}} = \frac{\operatorname{sech}^2 x}{\sqrt{\tanh^2 x-1}}$

18) c)

$y = \operatorname{artanh}(\sinh x)$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \cosh x \times \frac{1}{1-\sinh^2 x} = \frac{\cosh x}{1-\sinh^2 x}$

18) d)

$y = \operatorname{arsinh}(\tanh x)$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \operatorname{sech}^2 x \times \frac{1}{\sqrt{\tanh^2 x + 1}} = \frac{\operatorname{sech}^2 x}{\sqrt{\tanh^2 x + 1}}$

18) e)

$y = \operatorname{arcosh}(\sinh x)$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \cosh x \times \frac{1}{\sqrt{\sinh^2 x - 1}} = \frac{\cosh x}{\sqrt{\sinh^2 x - 1}}$

18) f)

$y = \operatorname{artanh}(\cosh x)$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \sinh x \times \frac{1}{1 - \cosh^2 x} = \frac{\sinh x}{1 - \cosh^2 x}$

19) a)

$y = x^2 \operatorname{arsinh} x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = x^2 \times \frac{1}{\sqrt{x^2 + 1}} + 2x \times \operatorname{arsinh} x$

$= \frac{x^2}{\sqrt{x^2 + 1}} + 2x \operatorname{arsinh} x$

19) b)

$y = \sinh x \operatorname{arsinh} x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \sinh x \times \frac{1}{\sqrt{x^2 + 1}} + \cosh x \times \operatorname{arsinh} x$

$= \frac{\sinh x}{\sqrt{x^2 + 1}} + \cosh x \operatorname{arsinh} x$

19) c)

$y = \cosh x \operatorname{arcosh} x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \cosh x \times \frac{1}{\sqrt{x^2 - 1}} + \sinh x \times \operatorname{arcosh} x$

$= \frac{\cosh x}{\sqrt{x^2 - 1}} + \sinh x \operatorname{arcosh} x$

19) d)

$y = \operatorname{arsinh} x \operatorname{arcosh} x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \operatorname{arsinh} x \times \frac{1}{\sqrt{x^2-1}} + \frac{1}{\sqrt{x^2+1}} \times \operatorname{arcosh} x$

$= \frac{\operatorname{arsinh} x}{\sqrt{x^2-1}} + \frac{\operatorname{arcosh} x}{\sqrt{x^2+1}}$

19) e)

$y = \operatorname{artanh} 2x \operatorname{artanh} 4x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \operatorname{artanh} 2x \times 4 \times \frac{1}{1-(4x)^2} + 2 \times \frac{1}{1-(2x)^2} \times \operatorname{artanh} 4x$

$= \frac{4 \operatorname{artanh} 2x}{1-16x^2} + \frac{2 \operatorname{artanh} 4x}{1-4x^2}$

20) a)

$y = \frac{x}{\operatorname{arsinh} x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\operatorname{arsinh} x \times 1 - x \times \frac{1}{\sqrt{x^2+1}}}{(\operatorname{arsinh} x)^2}$

$= \frac{\operatorname{arsinh} x - \frac{x}{\sqrt{x^2+1}}}{(\operatorname{arsinh} x)^2}$

20) b)

$y = \frac{\operatorname{arsinh} x}{\operatorname{arcosh} x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\operatorname{arcosh} x \times \frac{1}{\sqrt{x^2+1}} - \operatorname{arsinh} x \times \frac{1}{\sqrt{x^2-1}}}{(\operatorname{arcosh} x)^2}$

$= \frac{\frac{\operatorname{arcosh} x}{\sqrt{x^2+1}} - \frac{\operatorname{arsinh} x}{\sqrt{x^2-1}}}{(\operatorname{arcosh} x)^2}$

20) c)

$y = \frac{\operatorname{arcosh} 3x}{\operatorname{arsinh} 6x}$

$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\operatorname{arsinh} 6x \times 3 \times \frac{1}{\sqrt{(3x)^2-1}} - \operatorname{arcosh} 3x \times 6 \times \frac{1}{\sqrt{(6x)^2+1}}}{(\operatorname{arsinh} 6x)^2}$

$= \frac{\frac{3 \operatorname{arsinh} 6x}{\sqrt{9x^2-1}} - \frac{6 \operatorname{arcosh} 3x}{\sqrt{36x^2+1}}}{(\operatorname{arsinh} 6x)^2}$

20) d)

$$y = \frac{\operatorname{arsinh} 4x}{\operatorname{artanh} 8x}$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\operatorname{artanh} 8x \times 4 \times \frac{1}{\sqrt{(4x)^2+1}} - \operatorname{arsinh} 4x \times 8 \times \frac{1}{1-(8x)^2}}{(\operatorname{artanh} 8x)^2}$$

$$= \frac{\frac{4 \operatorname{artanh} 8x}{\sqrt{16x^2+1}} - \frac{8 \operatorname{arsinh} 4x}{1-64x^2}}{(\operatorname{artanh} 8x)^2}$$

21)

$$y = \sinh(2 \operatorname{arcosh} x)$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{2}{\sqrt{x^2-1}} \times \cosh(2 \operatorname{arcosh} x) = \frac{2 \cosh(2 \operatorname{arcosh} x)}{\sqrt{x^2-1}}$$

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{\sqrt{x^2-1} \times \frac{2}{\sqrt{x^2-1}} \times 2 \sinh(2 \operatorname{arcosh} x) - 2 \cosh(2 \operatorname{arcosh} x) \times x(x^2-1)^{-\frac{1}{2}}}{\left(\sqrt{x^2-1}\right)^2}$$

$$= \frac{4 \sinh(2 \operatorname{arcosh} x) - \frac{2x \cosh(2 \operatorname{arcosh} x)}{\sqrt{x^2-1}}}{x^2-1}$$

$$= \frac{4\sqrt{x^2-1} \sinh(2 \operatorname{arcosh} x) - 2x \cosh(2 \operatorname{arcosh} x)}{(x^2-1)^{\frac{3}{2}}}$$

Now,

$$(x^2-1)\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + x\frac{\mathrm{d}y}{\mathrm{d}x}$$

$$= (x^2-1) \times \frac{4\sqrt{x^2-1} \sinh(2 \operatorname{arcosh} x) - 2x \cosh(2 \operatorname{arcosh} x)}{(x^2-1)^{\frac{3}{2}}} + x \times \frac{2 \cosh(2 \operatorname{arcosh} x)}{\sqrt{x^2-1}}$$

$$= \frac{4\sqrt{x^2-1} \sinh(2 \operatorname{arcosh} x) - 2x \cosh(2 \operatorname{arcosh} x)}{\sqrt{x^2-1}} + \frac{2x \cosh(2 \operatorname{arcosh} x)}{\sqrt{x^2-1}}$$

$$= 4 \sinh(2 \operatorname{arcosh} x) - \frac{2x \cosh(2 \operatorname{arcosh} x)}{\sqrt{x^2-1}} + \frac{2x \cosh(2 \operatorname{arcosh} x)}{\sqrt{x^2-1}}$$

$$= 4 \sinh(2 \operatorname{arcosh} x)$$

$$= 4y \text{ as required}$$

22)

$y = (\operatorname{arsinh} 4x)^2$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 4 \times 2 \times \frac{1}{\sqrt{(4x)^2+1}} \times \operatorname{arsinh} 4x$

$= \frac{8 \operatorname{arsinh} 4x}{\sqrt{16x^2+1}}$

$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{\sqrt{16x^2+1} \times 8 \times 4 \times \frac{1}{\sqrt{(4x)^2+1}} - 8 \operatorname{arsinh} 4x \times 32x \times \frac{1}{2}(16x^2+1)^{-\frac{1}{2}}}{\left(\sqrt{16x^2+1}\right)^2}$

$= \frac{32 - \frac{128x \operatorname{arsinh} 4x}{\sqrt{16x^2+1}}}{16x^2+1}$

$= \frac{32\sqrt{16x^2+1} - 128x \operatorname{arsinh} 4x}{(16x^2+1)^{\frac{3}{2}}}$

Now,

$(16x^2+1)\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + 16x\frac{\mathrm{d}y}{\mathrm{d}x}$

$= (16x^2+1) \times \frac{32\sqrt{16x^2+1} - 128x \operatorname{arsinh} 4x}{(16x^2+1)^{\frac{3}{2}}} + 16x \times \frac{8 \operatorname{arsinh} 4x}{\sqrt{16x^2+1}}$

$= \frac{32\sqrt{16x^2+1} - 128x \operatorname{arsinh} 4x}{\sqrt{16x^2+1}} + \frac{128x \operatorname{arsinh} 4x}{\sqrt{16x^2+1}}$

$= \frac{32\sqrt{16x^2+1}}{\sqrt{16x^2+1}} - \frac{128x \operatorname{arsinh} 4x}{\sqrt{16x^2+1}} + \frac{128x \operatorname{arsinh} 4x}{\sqrt{16x^2+1}}$

$= 32$

23)

Let $y = \operatorname{arcosech} x$

Then $\operatorname{cosech} y = x$

$\frac{\mathrm{d}}{\mathrm{d}x}(\operatorname{cosech} y) = \frac{\mathrm{d}}{\mathrm{d}x}(x)$

$\Rightarrow -\coth y \operatorname{cosech} y \frac{\mathrm{d}y}{\mathrm{d}x} = 1$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{\coth y \operatorname{cosech} y} = -\frac{1}{\sqrt{\operatorname{cosech}^2 y + 1} \times \operatorname{cosech} y} = -\frac{1}{x\sqrt{x^2+1}}$ as required

24)

Let $y = \operatorname{arsech} x$

Then $\operatorname{sech} y = x$

$\frac{\mathrm{d}}{\mathrm{d}x}(\operatorname{sech} y) = \frac{\mathrm{d}}{\mathrm{d}x}(x)$

$\Rightarrow -\tanh y \operatorname{sech} y \frac{\mathrm{d}y}{\mathrm{d}x} = 1$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{\tanh y \operatorname{sech} y} = -\frac{1}{\sqrt{1-\operatorname{sech}^2 y} \times \operatorname{sech} y} = -\frac{1}{x\sqrt{1-x^2}}$ as required

25)

Let $y = \operatorname{arcoth} x$

Then $\coth y = x$

$\frac{\mathrm{d}}{\mathrm{d}x}(\coth y) = \frac{\mathrm{d}}{\mathrm{d}x}(x)$

$\Rightarrow -\operatorname{cosech}^2 y \frac{\mathrm{d}y}{\mathrm{d}x} = 1$

$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{1}{\operatorname{cosech}^2 y} = -\frac{1}{\coth^2 y - 1} = -\frac{1}{x^2-1} = \frac{1}{1-x^2}$ as required

26) a)

$y = x^2 \operatorname{arcosech} 2x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = x^2 \times 2 \times -\frac{1}{(2x)\sqrt{(2x)^2+1}} + 2x \operatorname{arcosech} 2x$

$= -\frac{2x^2}{2x\sqrt{4x^2+1}} + 2x \operatorname{arcosech} 2x$

$= -\frac{x}{\sqrt{4x^2+1}} + 2x \operatorname{arcosech} 2x$

26) b)

$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=\frac{1}{2}} = -\frac{\frac{1}{2}}{\sqrt{4\left(\frac{1}{2}\right)^2+1}} + 2\left(\frac{1}{2}\right) \operatorname{arcosech} 2\left(\frac{1}{2}\right)$

$= -\frac{\sqrt{2}}{4} + \operatorname{arcosech} 1$

$= -\frac{\sqrt{2}}{4} + \ln\left(\frac{1}{1} + \sqrt{\frac{1}{1^2}+1}\right)$

$= \ln\left(1+\sqrt{2}\right) - \frac{\sqrt{2}}{4}$

27)

$y = \operatorname{arsech} x^2 + \operatorname{arcoth} 3x$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 2x \times -\frac{1}{(x^2)\sqrt{1-(x^2)^2}} + \frac{3}{1-(3x)^2}$

$= -\frac{2}{x\sqrt{1-x^4}} + \frac{3}{1-9x^2}$

$\left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=-\frac{1}{2}} = -\frac{2}{\left(-\frac{1}{2}\right)\sqrt{1-\left(-\frac{1}{2}\right)^4}} + \frac{3}{1-9\left(-\frac{1}{2}\right)^2}$

$= \frac{4}{\sqrt{\frac{15}{16}}} - \frac{12}{5}$

$= \frac{16\sqrt{15}}{15} - \frac{12}{5}$